SIOP Reporting Case Study – Documentation

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1. Understanding of the Case Study

The objective of this case study was to develop an interactive Sales, Inventory & Operations Planning (SIOP) reporting dashboard using Power BI. The dataset consisted of two primary sources:

- **Forecast Data:** Multiple monthly forecast snapshots (As Of Periods) containing Revenue, Non-Revenue, and CAPEX forecasts.
- **Actual Data:** Actual realized Revenue, Non-Revenue, and CAPEX quantities by Country and Period.

The primary business requirement was to compare multiple forecast snapshots for the same future period. For the latest available snapshot, forecast values were labeled as **Resultant**, while the immediately preceding snapshots were labeled **Lag 0** and **Lag 1**. This allows planners to understand how forecasts evolve over successive planning cycles.

In addition, the dashboard was required to calculate Forecast Accuracy and Forecast Bias KPIs, provide country-level executive reporting, and enable users to dynamically switch between forecast snapshots.

2. Solution Approach

Data Preparation

The forecast and actual datasets were imported into Power BI and transformed using Power Query.

The following preparation steps were performed:

- Cleaned and validated both datasets.
- Created a **Snapshot Reference** table to identify the latest forecast snapshot and assign corresponding Lag values (Resultant, Lag 0, Lag 1).
- Created a **CountryPeriod** key to associate forecast and actual records.
- Built supporting dimension tables to establish appropriate relationships between forecast and actual data.

Forecast Transformation

The forecast dataset was transformed to support snapshot comparison by assigning each forecast record to one of three categories:

- Resultant
- Lag 0
- Lag 1

Dynamic DAX measures were created to retrieve the selected forecast values based on the user's Lag selection.

This approach enables the dashboard to remain scalable as additional forecast snapshots are introduced.

KPI Development

Business KPIs were implemented according to the requirements.

Forecast Accuracy %

Forecast Accuracy measures how closely forecast values match actual values.

Formula used:

Forecast Accuracy = 1 − ABS(Forecast − Actual) / Actual

Forecast Bias %

Forecast Bias measures whether forecasts systematically overestimate or underestimate actual performance.

Formula used:

Forecast Bias = (Forecast − Actual) / Actual

Positive values indicate over-forecasting, while negative values indicate under-forecasting.

Dashboard Development

An executive dashboard was designed to provide both high-level KPIs and detailed analytical views.

The dashboard includes:

- Lag Selection slicer (Resultant, Lag 0, Lag 1)

- Period selection slicer
- Revenue Forecast KPI
- Non-Revenue Forecast KPI
- CAPEX Forecast KPI
- Actual Revenue KPI
- Forecast Accuracy KPI
- Forecast Bias KPI
- Revenue Forecast by Country
- Non-Revenue Forecast by Country
- CAPEX Forecast by Country

Additional Analytical Enhancements

To provide deeper business insights beyond the original requirements, several analytical features were incorporated.

Forecast Revision Analysis

Additional measures were developed to compare Resultant forecasts with previous snapshots.

Revenue variance between Resultant and Lag 0 was calculated to identify forecast revisions across planning cycles.

Forecast Evolution

A clustered column chart was created to compare Resultant and Lag 0 forecasts across planning periods, allowing users to visualize how forecasts changed over time.

Forecast Accuracy Trend

A line chart was developed to display Forecast Accuracy across planning periods, helping identify periods with stronger or weaker forecasting performance.

Bias Status Indicator

Forecast Bias values were categorized using predefined thresholds to quickly identify acceptable forecasts and critical deviations, making the dashboard easier for executive interpretation.

3. Key Insights

The dashboard enables planners and business users to:

- Compare forecast revisions across multiple planning snapshots.
- Measure forecast quality using Forecast Accuracy and Forecast Bias.
- Identify countries with significant forecast deviations.
- Monitor Revenue, Non-Revenue, and CAPEX forecasts simultaneously.
- Analyze forecast evolution across planning periods.
- Support data-driven decision-making during monthly SIOP review meetings.

4. Scalability

The solution has been designed to be scalable and reusable.

Since Lag assignments are dynamically generated from forecast snapshots, the dashboard can accommodate future planning cycles without structural modifications. New forecast snapshots can be incorporated by refreshing the dataset, enabling continuous monitoring of forecast performance.

5. Conclusion

The developed Power BI solution satisfies the functional requirements of the case study while providing additional analytical capabilities that enhance decision-making. By combining dynamic lag comparison, executive KPIs, country-level analysis, and forecast trend visualization, the dashboard provides a comprehensive view of forecasting performance and supports continuous improvement in the SIOP planning process.